

Chapter 13 — Incisions, Exposure, and Drains — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Fascial bites** 1 cm from edge, 1 cm apart
- **Continuous + rapidly absorbable suture** → ↑ hernia
- **Gibson** = **standard** for renal transplant and transplant nephrectomy
- **Thoracoabdominal** → level III caval thrombus, T4, large upper-pole/adrenal mass
- **Flank** → **pleural risk**; **anterior** → **bowel risk**, early vascular control
- **Chest radiograph** only if **pleural tear repaired** or suspected
- **Scrotal incision**: only **contraindication** = **suspected testicular malignancy** → **inguinal**
- **Radical orchiectomy**: 5–7 cm, from 2 cm cephalad/lateral to pubic tubercle
- **Subinguinal ILND** (inguinal lymph node dissection) 2 cm below ligament; commonest; flap necrosis, lymphocele
- **Drains**: **no evidence for routine use**; closed suction, Penrose only scrotum

ABDOMINAL INCISIONS

- **Midline**: most versatile; incise **linea alba** → avoids rectus vessels
- **Lower midline** = commonest urological incision (bladder, prostate)
- **Complete midline**: trauma nephrectomy, RPLND (retroperitoneal lymph node dissection), polycystic bilateral nephrectomy
- **Closure**: one layer; running = interrupted for hernia
- **Smead-Jones**: near-far figure-of-eight, 2 cm apart
- **Pfannenstiel**: 2 fingerbreadths above pubis; ↓ hernia vs midline; extraction site
- **Gibson**: medial to ASIS (anterior superior iliac spine) → just lateral to rectus; extraperitoneal
- **Gibson**: ureter found crossing iliac artery

ANTERIOR APPROACHES TO KIDNEY

- **Transverse** → paediatric Wilms, pyeloplasty
- **Paramedian** → rectus nerve injury, ventral hernia; avoid unless complex midline
- **Midline transperitoneal** → trauma, IVC (inferior vena cava) thrombus, horseshoe kidney
- **Chevron** → bilateral tumours, left-sided caval thrombus; liver/spleen injury
- **Subcostal**: 2 fingerbreadths below costal margin → tip of 11th rib
- **Supine subcostal** → better hilar control for large anterior tumours
- **Subcostal + xiphoid** → suprahepatic IVC; + **sternotomy** → atrial thrombus
- **Thoracoabdominal**: 9th–10th rib bed; modified via 8th rib bed
- **Chest tube** omissible if parietal pleura intact
- **Diaphragm**: two layers, running absorbable

FLANK INCISIONS

- **Flank simple nephrectomy** → XGP (xanthogranulomatous pyelonephritis)
- **Layers**: external oblique → internal oblique → transversus abdominis
- **Position**: lower leg flexed, upper extended, over table break
- **Supracostal** → spares rib, pleural risk
- **11th transcostal** → pleural injury + **flank bulge**
- **Doyen rib stripper** frees rib posteriorly; bone wax to edges
- **Neurovascular bundle** between layers 2 and 3
- **Pleural tear**: red rubber catheter under water, big breaths, remove when no bubbles
- **Large pleural tear** → formal chest tube

- **Reoperation**: extend into unviolated tissue, enter peritoneum there
- **General surgeon** → prior cystectomy/diversion, trauma, bowel surgery

INGUINAL INCISIONS

- **Benign inguinal** (varicocele, orchidopexy): 2–3 cm, 2 fingerbreadths above symphysis
- **Aim**: just below external ring
- **Subinguinal** → obese or prior inguinal surgery
- **Protect ilioinguinal nerve**; new external ring not too tight
- **ILND flaps**: below Scarpa fascia; ± myocutaneous flap
- **Nodal extent** chooses ILND/PLND (pelvic lymph node dissection) incision

SPECIFIC SURGERIES

- **Bilateral scrotal** → vertical median raphe
- **Scrotum**: close dead space or Penrose → prevent haematoma
- **Adult circumcision**: two-incision sleeve; commonest
- **Inner incision** 5–10 mm from glans; outer at coronal sulcus
- **Frenular vessels** ventrally — highest bleeding risk
- **Closure**: 3-0/4-0 chromic interrupted; continuous → strangulation risk
- **Penile fracture**: degloving standard; direct incision → better sexual function, needs imaging
- **Perineal urethral**: 8–10 cm vertical from perineoscrotal junction
- **Lambda (inverted-Y)** → ↑ 90-day wound complications
- **Colles fascia** closed before skin
- **Radical perineal prostatectomy**: inverted horseshoe, apex ~2 cm from anal verge
- **Balfour 1912** → retropubic, transvesical
- **Finochietto 1941** → rib spreader, chest approach
- **Bookwalter/Omni-Tract/Thompson** → table-fixed; extra articulation no benefit
- **Lone Star** (F. Brantley Scott) → perineal, stepwise layers
- **Retractor safety**: moist pads first, minimum force
- **Drain**: separate stab incision; avoid inferior epigastrics; nonabsorbable stitch
- **Drain if**: partial nephrectomy, XGP nephrectomy

CLASSIC TRAPS

- **Trap**: thoracoabdominal more painful than flank — **Actually**: no difference
- **Trap**: routine chest radiograph after nephrectomy — **Actually**: only if tear repaired
- **Trap**: scrotal approach to testicular mass — **Actually**: inguinal

- **Trap:** hockey-stick Gibson safer — **Actually:** higher hernia rates
- **Trap:** wound protectors cut infection in cystectomy — **Actually:** general surgery only
- **Trap:** transcostal spares rib — **Actually:** supracostal spares rib
- **Trap:** Lambda perineal is superior — **Actually:** more wound complications

- **Trap:** routine drains after RARP (robotic-assisted radical prostatectomy) or partial nephrectomy — **Actually:** little evidence; discretion
- **Trap:** Penrose default drain — **Actually:** closed suction; Penrose for scrotum
- **Trap:** paramedian as safe midline alternative — **Actually:** nerve injury, ventral hernia